CAMBRIDGE
International Education

Cambridge IGCSE™

DESIGN & TECHNOLOGY **0445/13**

Paper 1 Product Design **May/June 2026**

1 hour 15 minutes

You must answer on the two pre-printed A3 answer sheets.

You will need: Two A3 pre-printed answer sheets (enclosed)
Standard drawing equipment
Coloured pencils

INSTRUCTIONS
- Answer **one** question.
- Use an HB pencil for any drawings and a black or dark blue pen for any writing.
- Write your name, centre number and candidate number in the space on **both** pre-printed answer sheets.
- Answer in the space provided on the answer sheets.
- Do **not** use an erasable pen, staples, paper clips, glue, or correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.
- You may use standard drawing equipment, including coloured pencils.
- At the end of the examination, hand in your named A3 answer sheets. Do **not** fasten them together and do **not** punch holes in the sheets or tie with string.

INFORMATION
- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [ ].
- All dimensions are in millimetres unless otherwise stated.

This document has **4** pages.

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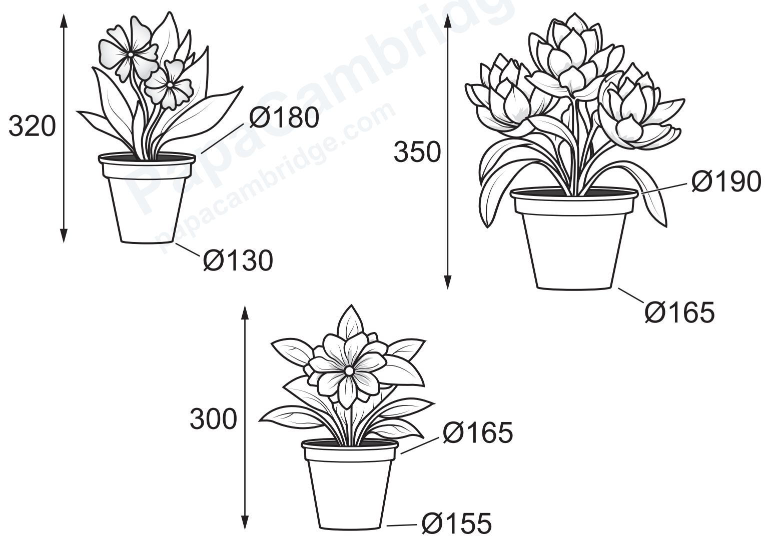
[Turn over

Answer **one** question only on the A3 pre-printed answer sheets provided.

1 Potted household plants are often displayed on stands.



Design a floor standing unit that will display the three potted household plants shown.
The unit must be adjustable for different heights.

(a) List four additional points about the function of such a floor standing unit that you consider to be important. [4]

(b) Use sketches and notes to show **two** methods of adjusting the height of products. [4]

(c) Develop and sketch **three** separate ideas for the floor standing unit. [12]

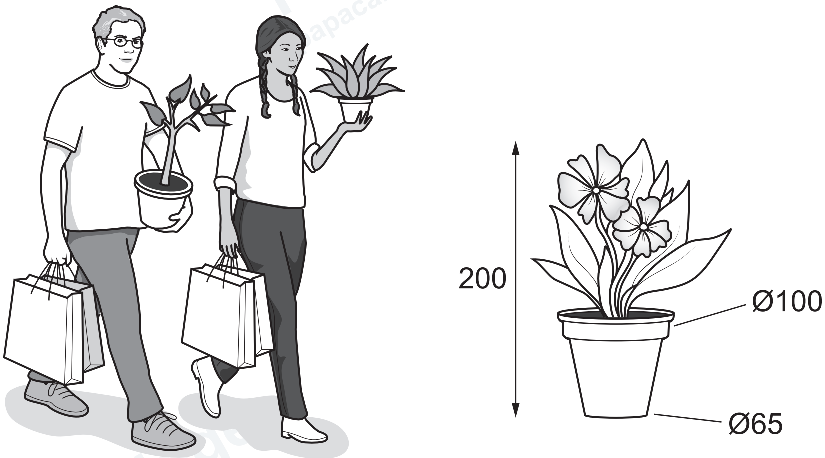
(d) Evaluate your three ideas. Choose **one** idea to develop further and justify your choice. [8]

(e) Draw, using a method of your own choice, a full solution to the design problem. Include construction details and important dimensions. [12]

(f) Suggest **two** suitable specific materials for the solution you have drawn in **1(e)** and give reasons for your choice. [4]

(g) Outline a method that could be used to manufacture **one** part of your solution drawn in **1(e)**. Include the names of the tools used. [6]

2 Bought potted plants need transporting home.



Design a product that will allow four potted plants, of the size shown, to be carried in one hand. The product must be flat-packed prior to use and made from lightweight graphic materials.

(a) List **four** additional points about the function of such a product that you consider important. [4]

(b) Use sketches and notes to show **two** methods of joining lightweight graphic materials to make three dimensional (3D) shapes without the use of adhesives. [4]

(c) Develop and sketch **three** separate ideas for the product. [12]

(d) Evaluate your three ideas. Choose **one** idea to develop further and justify your choice. [8]

(e) Draw, using a method of your own choice, a full solution to the design problem. Include construction details and important dimensions. [12]

(f) Suggest **two** suitable specific materials for the solution you have drawn in **2(e)** and give reasons for your choice. [4]

(g) Outline a method that could be used to manufacture **one** part of your solution drawn in **2(e)**. Include the names of the tools used. [6]

3 Soil and granular fertiliser are added to plant pots to allow the plants to grow.

Design a portable device that will add a measured amount of granular fertiliser and garden soil into a plant pot. The mixture must be in a ratio of one part granular fertiliser to five parts soil.

(a) List **four** additional points about the function of such a device that you consider to be important. [4]

(b) Use sketches and notes to show **two** methods that will control the release of a measured amount of granular material. [4]

(c) Develop and sketch **three** separate ideas for the device. [12]

(d) Evaluate your three ideas. Choose **one** idea to develop further and justify your choice. [8]

(e) Draw, using a method of your own choice, a full solution to the design problem. Include construction details and important dimensions. [12]

(f) Suggest **two** suitable specific materials for the solution you have drawn in **3(e)** and give reasons for your choice. [4]

(g) Outline a method that could be used to manufacture **one** part of your solution drawn in **3(e)**. Include the names of the tools used. [6]

